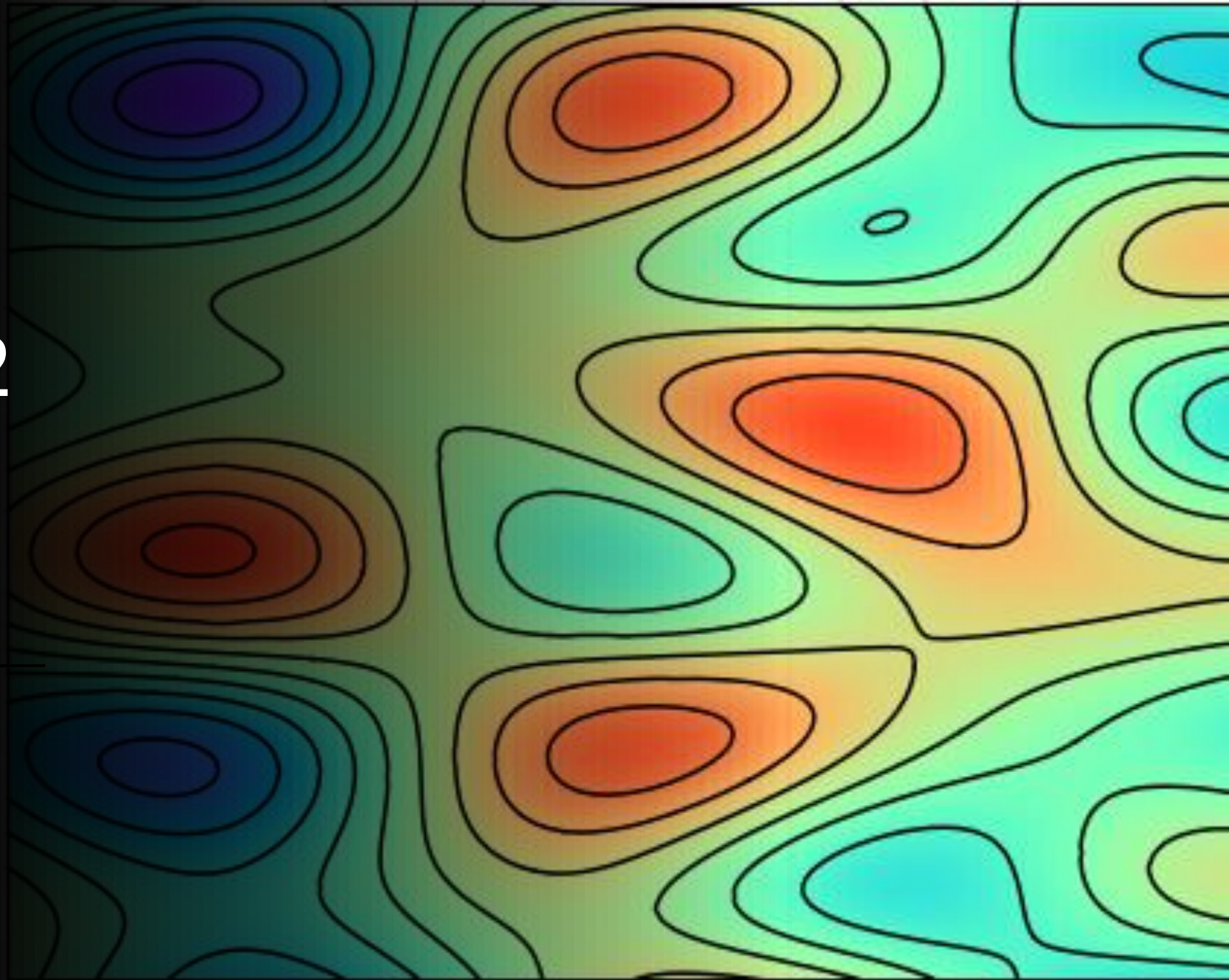




# Mathematics 2 (5EZB0) Lecture 3

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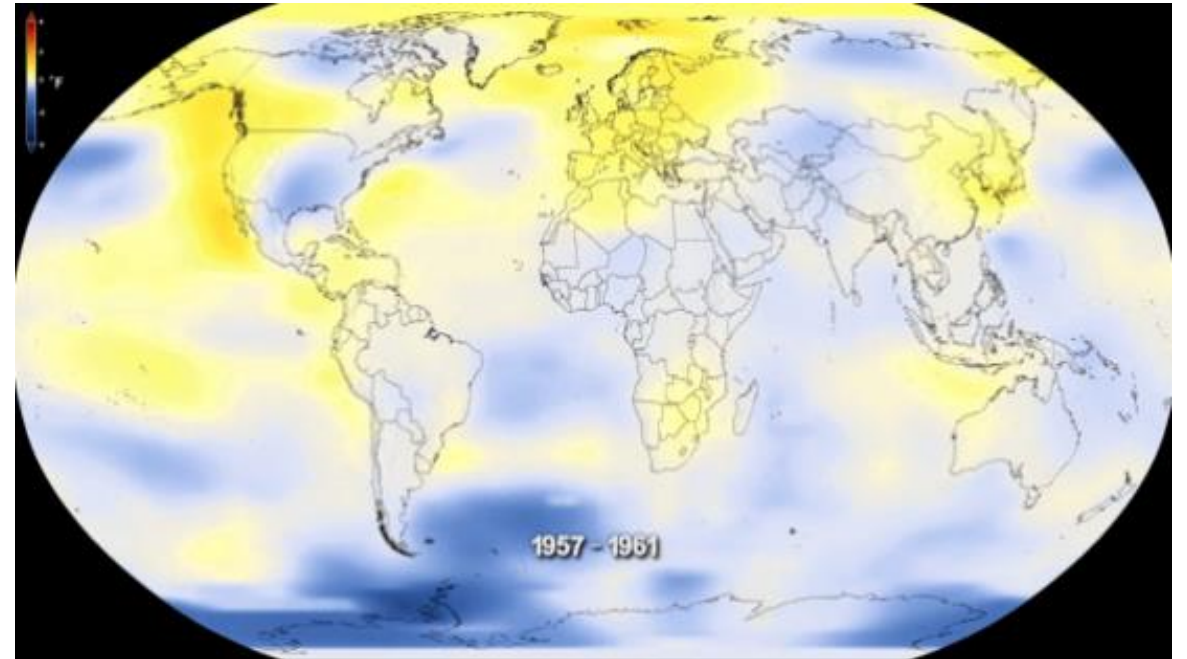
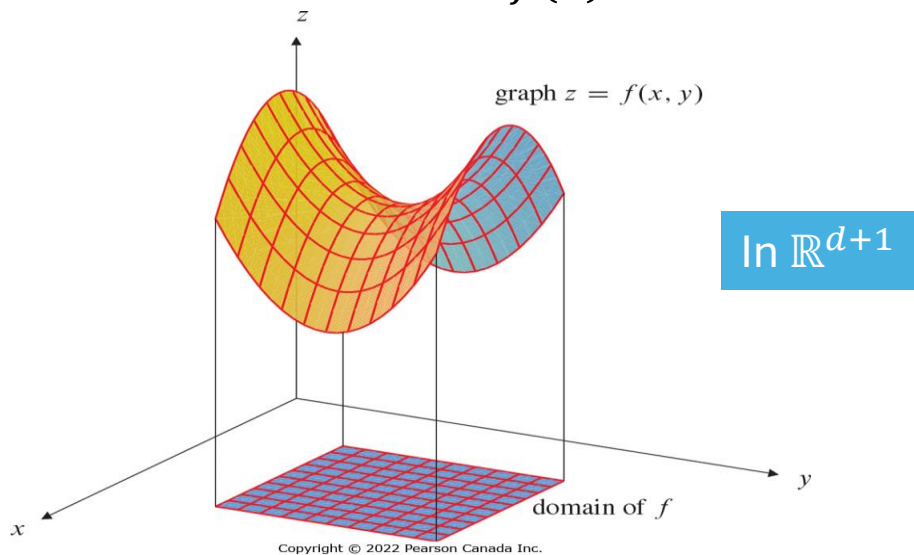
# Functions of several variables

Let  $\mathcal{D} \subseteq \mathbb{R}^d$ , be a *domain*. We denote a multivariate function  $f$  on the domain  $\mathcal{D}$  by

$$f: \mathcal{D} \rightarrow \mathbb{R}$$

also written for  $\mathbf{x} = (x_1, x_2, \dots, x_d) \in \mathcal{D}$  as

$$f(\mathbf{x})$$

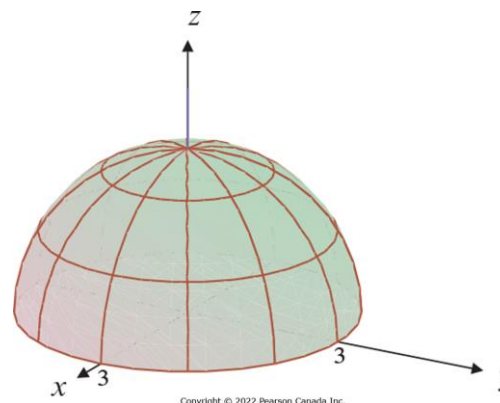
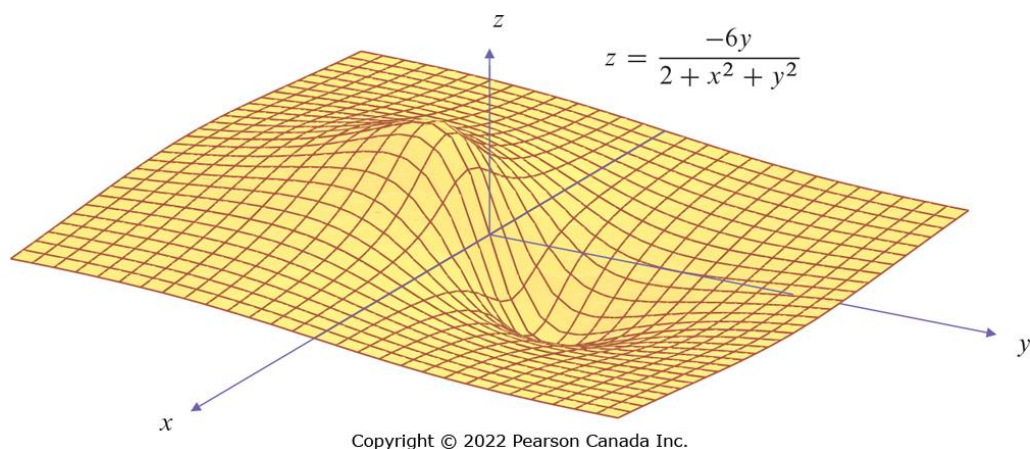


Source: Wikipedia

Use **graph3d.py** to define domains and functions in 3D to plot



# Domain & graph of a function



## Questions 1.2

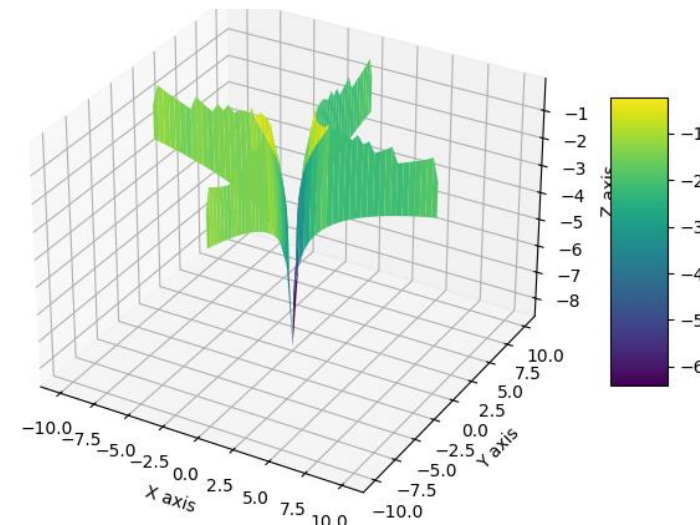
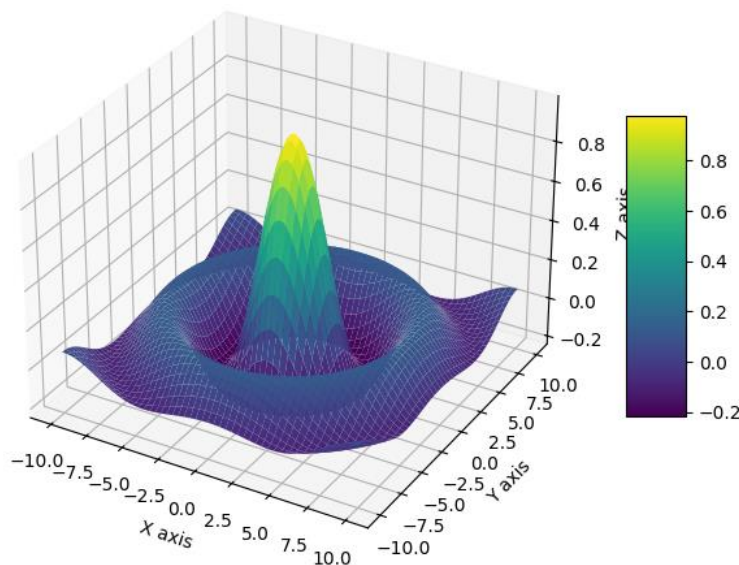
Can a sphere of radius 3 be represented as a graph in

1. Cartesian coordinates
2. Other coordinate systems

## Questions 1.1

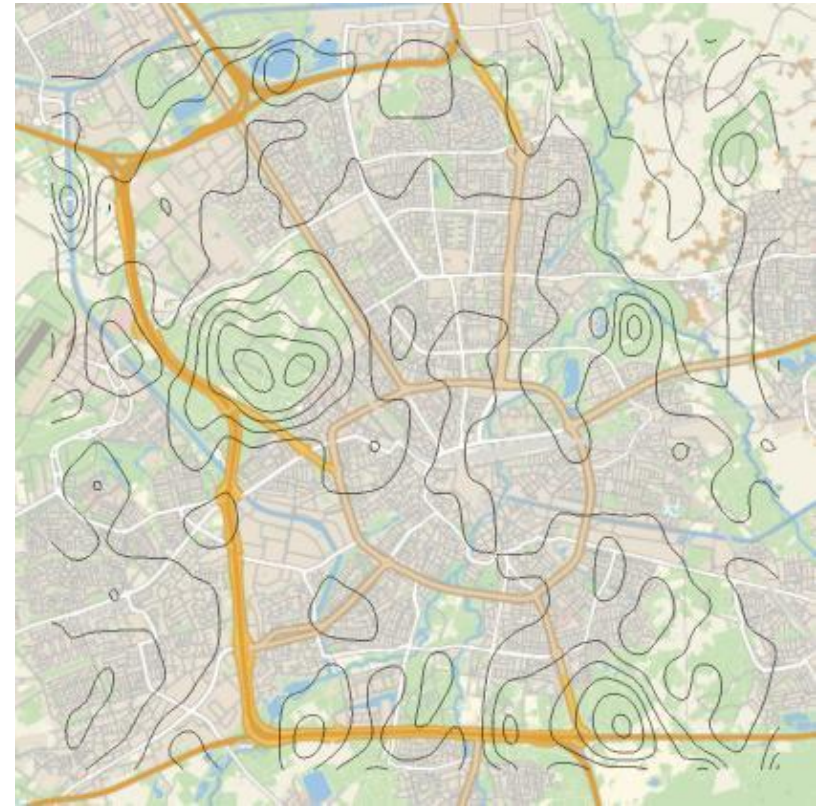
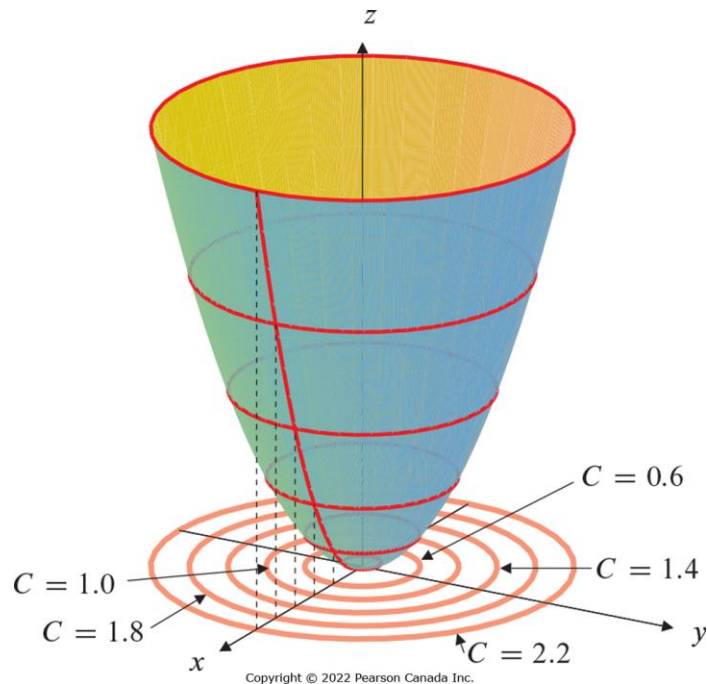
What is the maximum domain of definition of

1.  $\sin(\sqrt{x^2 + y^2})/\sqrt{x^2 + y^2}$
2.  $\ln(1 - \sqrt{1 - xy})$



# Level curves

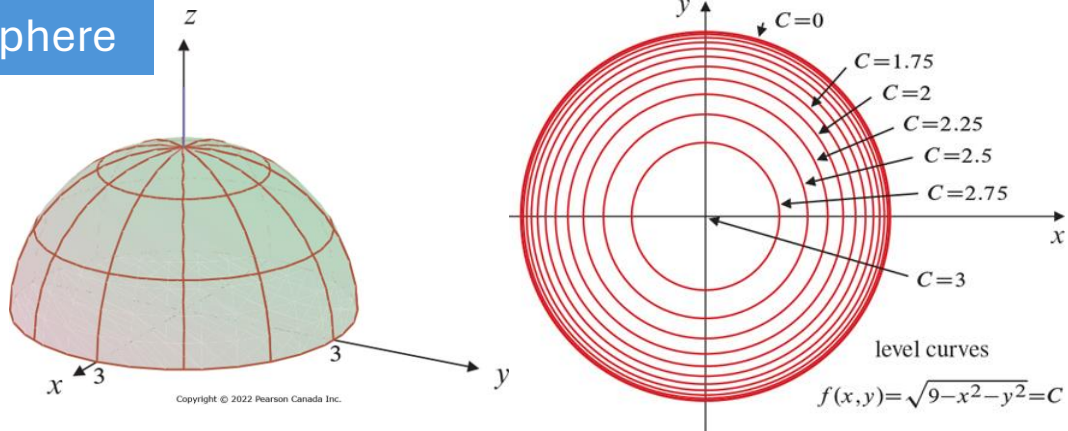
- Plots the contour of  $(x, y) \in \mathbb{R}^2$  for which  $f(x, y) = C$
- Two level curves cannot intersect
- Isolated maximum/minimum points are surrounded by closed loops



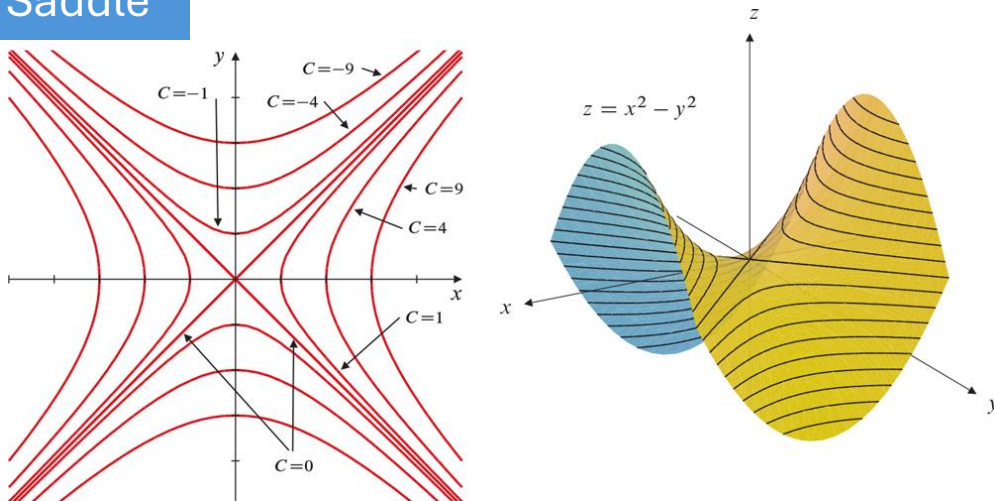
Heightmap of Eindhoven

# Level curves

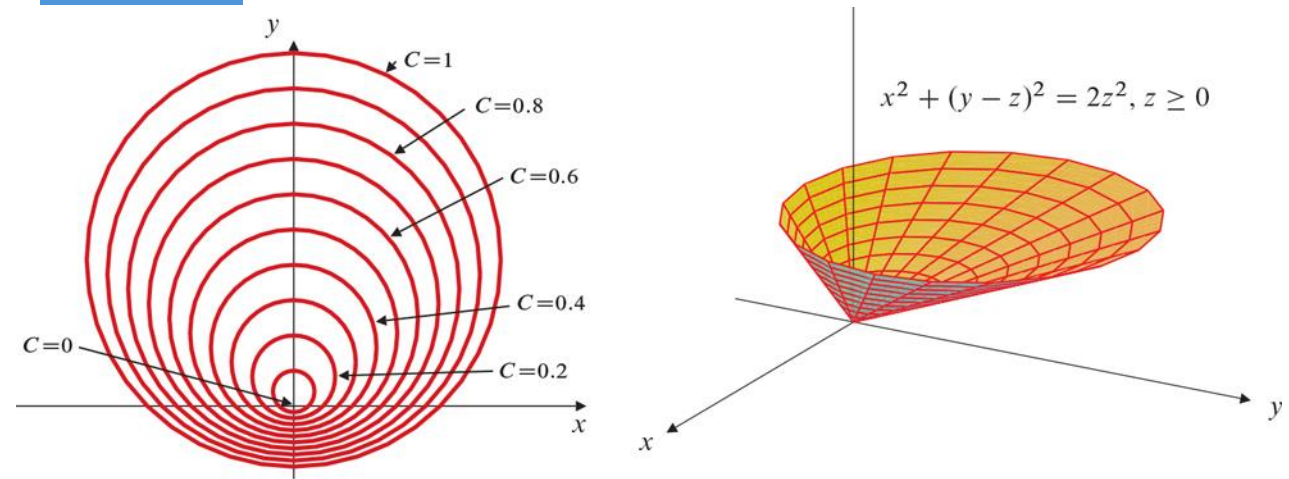
Sphere



Saddle



Cones

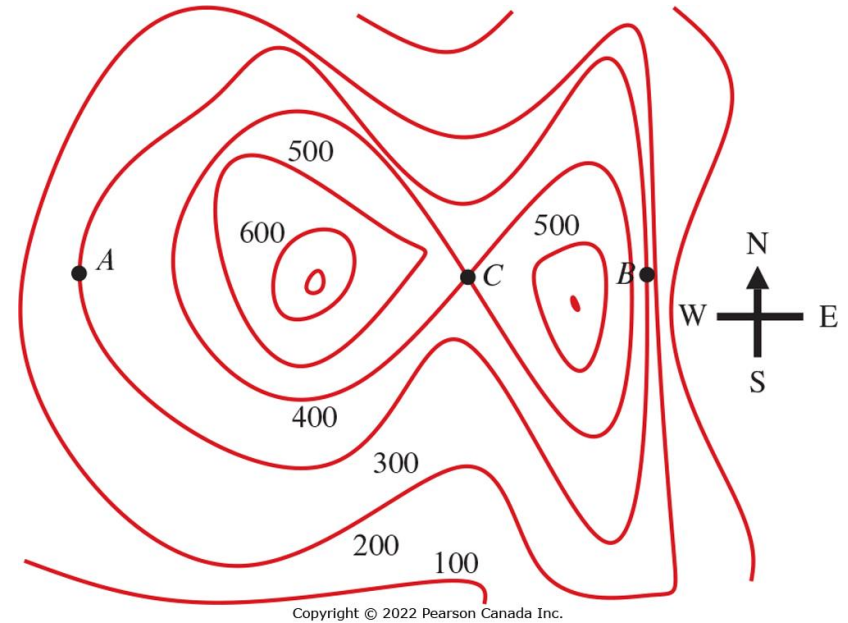




# Level curves

## Questions 2

1. Which point is steeper, A or B?
2. What is the topography of C?
3. Can you spot a peak?



# Limit

## Limit

$$\lim_{x \rightarrow x_0} f(x) = L$$

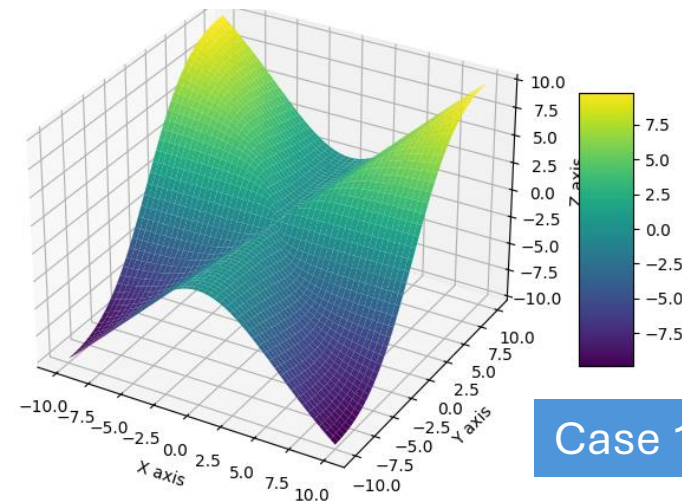
1. If  $x_0$  is not an isolated point, i.e., for all  $\delta > 0$   
 $B_\delta(x_0) \cap \mathcal{D} \neq \{x_0\}$
2. if for every  $\epsilon > 0$ ,  
there exists  $\delta > 0$ , such that  
 $|f(x) - L| < \epsilon$  for all  $x \in B_\delta(x_0) \cap \mathcal{D}$

➤ The limit, if existing, is unique

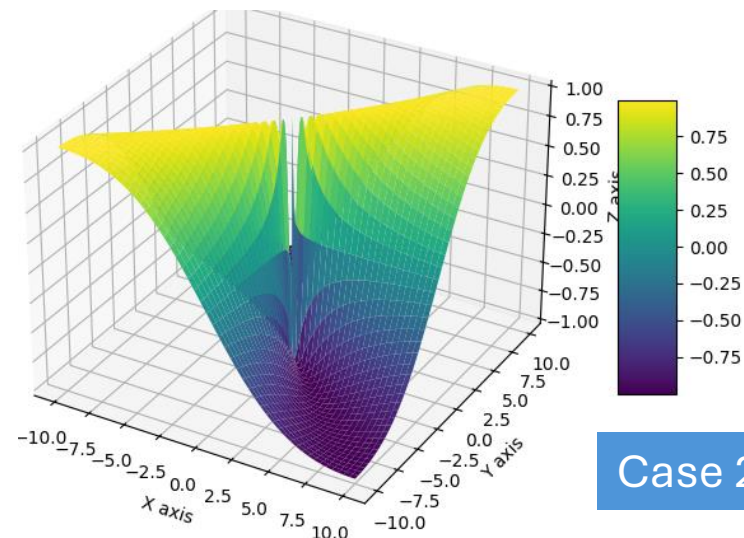
## Continuous functions

A function  $f$  is continuous at  $x = x_0$  if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$



Case 1:  $x^2y/(x^2 + y^2)$



Case 2:  $xy/(x^2 + y^2)$

# Limit

## Limit

$$\lim_{x \rightarrow x_0} f(x) = L$$

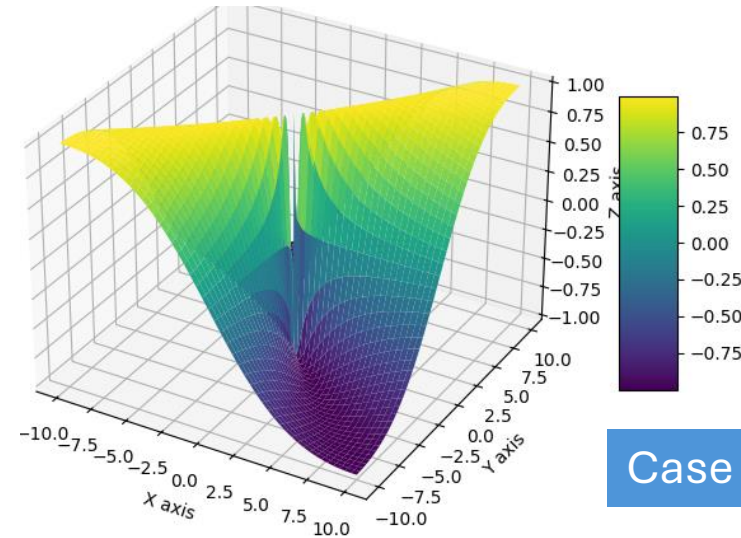
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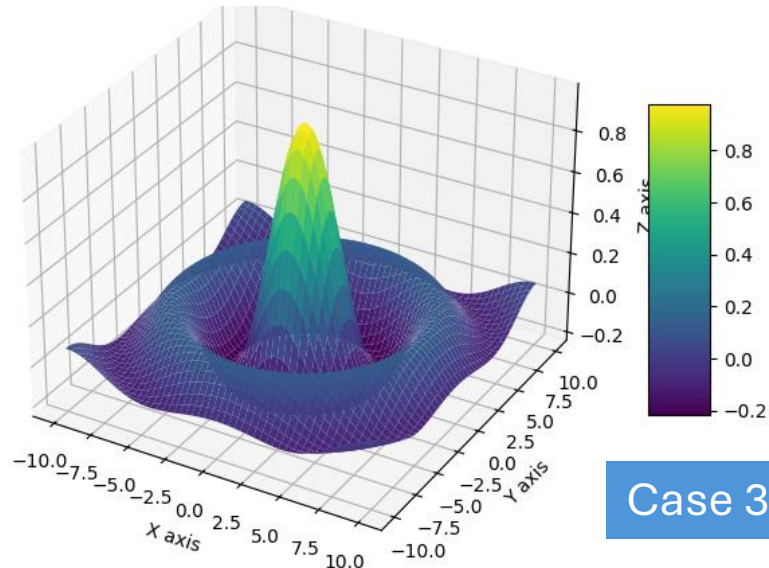
## Continuous functions

A function  $f$  is continuous at  $x = x_0$  if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$



Case 2:  $xy/(x^2 + y^2)$



Case 3:  $\sin r / r$

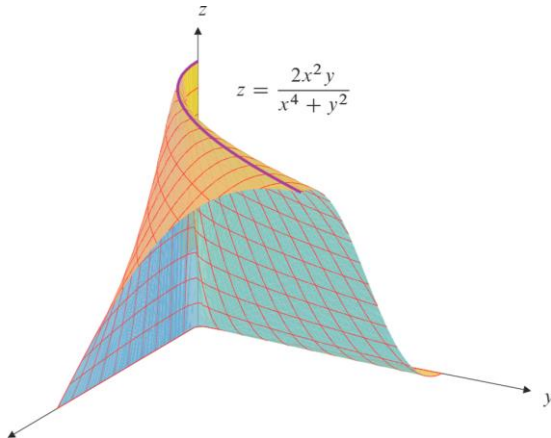


# Properties of limits

## Question 3.1

Do the following limits exist at  $(0,0)$ ?

1.  $\frac{x}{x^2+y^2}$
2.  $\frac{\sin x^2 y}{x^2+y^2}$
3.  $\frac{2x^2 y}{x^4+y^2}$



## Properties of limits

If  $x_0$  is not isolated in both the domains of  $f$  and  $g$ :

1.  $\lim_{x \rightarrow x_0} (f(x) \pm g(x)) = \lim_{x \rightarrow x_0} f(x) \pm \lim_{x \rightarrow x_0} g(x)$
2.  $\lim_{x \rightarrow x_0} (f(x)g(x)) = \lim_{x \rightarrow x_0} f(x) \lim_{x \rightarrow x_0} g(x)$
3.  $\lim_{x \rightarrow x_0} (f(x)/g(x)) = \lim_{x \rightarrow x_0} f(x) / \lim_{x \rightarrow x_0} g(x)$  if  $\lim_{x \rightarrow x_0} g(x) \neq 0$

Moreover, if  $F: \mathbb{R} \rightarrow \mathbb{R}$  is a continuous function, then

$$\lim_{x \rightarrow x_0} F(f(x)) = F(\lim_{x \rightarrow x_0} f(x))$$

## Question 3.2

Find the limit of  $\sin(x - y) / \cos(x + y)$  at  $(0,0)$ .

# Partial derivatives

Let the function  $f$  be continuous at a point  $(x, y) \in \mathcal{D}$ .

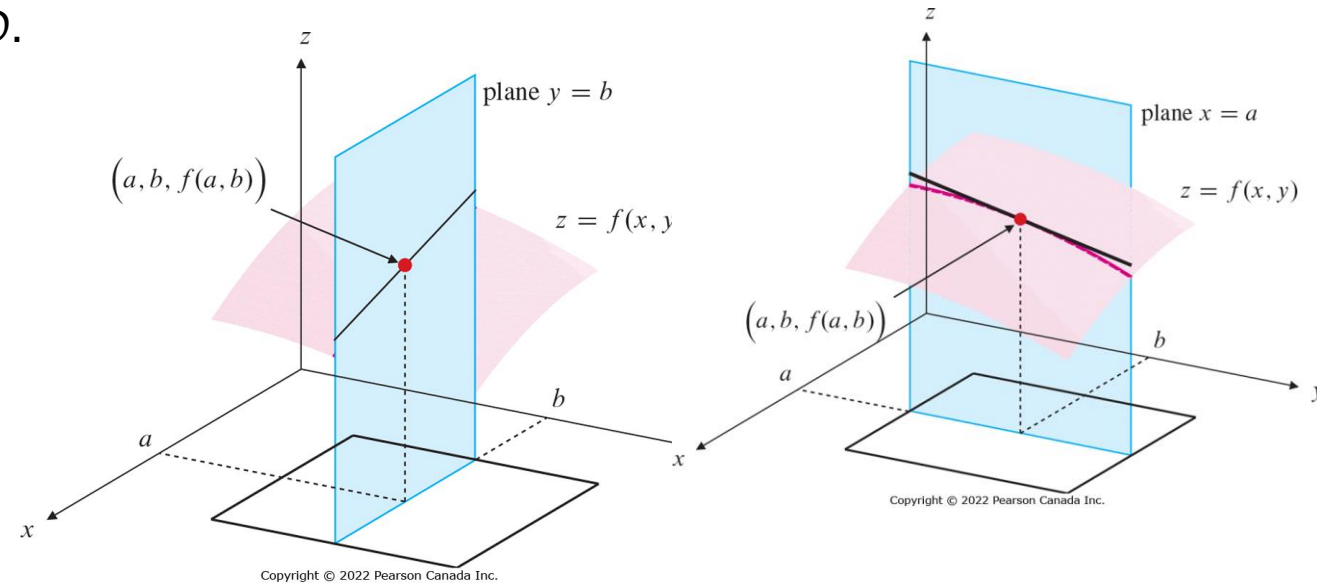
Its partial derivatives are then defined as

$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$\frac{\partial f}{\partial y}(x, y) = \lim_{k \rightarrow 0} \frac{f(x, y + k) - f(x, y)}{k}$$

If these limits exist.

- If you take partial derivative w.r.t one variable, then take all other variables to be constant, and use normal rules of differentiation
- $\frac{\partial f}{\partial x}$  is also commonly written as  $\partial_x f$ ,  $f_x$ , or  $f_1$



## Question 4

Find  $\partial_x f$  of the following functions

1.  $x^2 + xy + y^2$
2.  $e^{xy} \sin(x + y)$  at  $(0, \pi)$